

Exact Algorithms for Orthogonal and Unitary Tensor Equivalence over Finite Fields

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Abstract

When do two three-dimensional arrays describe the same tensor after orthogonal or unitary changes of coordinates? Over finite fields, this question cannot be approached through positivity or numerical diagonalization. We describe a construction that separates exact comparison from the probability of recognizing a suitable source. Two families of Gram contractions determine two coordinate changes up to scalars; equality of kernels and Gram matrices then decides whether the last change exists. A marked version handles the standard symmetric bilinear form in characteristic two. Small median dimension leads instead to bounded-variable polynomial equations, solved by finite component parameters and dimension-decreasing descent. A deterministic matrix-boundary algorithm completes the division into formats. We explain how primary-factor counts, a low-rank anchor estimate, and conditioned Gram laws give the proposed uniform average-case guarantee. The paper includes proofs of the central decision mechanisms, algorithm descriptions, and a source map for the longer counting and geometric arguments. The accompanying Lean development certifies substantial components, but its complete encoded-machine theorem remains open. We distinguish that unfinished certification from the mathematical construction and identify further algorithmic and probabilistic questions.

1 Introduction

An isomorphism problem asks whether different descriptions encode the same object. A tensor makes this question unusually demanding: changing coordinates in one direction changes every slice visible in the other two. Even a three-dimensional array can therefore conceal a large search over three interacting transformation groups. Restricting the transformations to isometries supplies additional information. Pairings of tensor entries survive a change of coordinates in a controlled form, and this information can be used to recover the transformations.

This paper develops that observation for tensors over finite fields. It is intended for readers comfortable with linear algebra who want to understand the complete route before entering its probability theory, algebraic geometry, or formal implementation. The organizing question is simple: which information do we retain at each reduction, and why is it enough to decide equivalence?

1.1 The problem and the meaning of average case

Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be positive integers. After permuting factors, assume $\mathbf{a} \leq \mathbf{b} \leq \mathbf{c}$. We write arrays in the order $\mathbf{a} \times \mathbf{c} \times \mathbf{b}$, putting the largest factor in the middle to simplify the contractions below. Thus $\mathbf{b}$, not the middle array index, denotes the median dimension throughout.

In the orthogonal case $K = \mathbb{F}_q$, set $\bar{\mathbf{x}} = \mathbf{x}$ and $\mathbf{X}^* = \mathbf{X}^T$. In the unitary case $K = \mathbb{F}_{q^2}$, set $\bar{\mathbf{x}} = \mathbf{x}^q$ and $\mathbf{X}^* = \bar{\mathbf{X}}^T$. In both cases write

$$\mathcal{G}_n(K) = \{P \in \text{Mat}_n(K) : P^*P = I_n\}. \quad (1.1)$$

The parameter q is the field size in the orthogonal case and the fixed-field size in the unitary case. Two arrays A, B are equivalent if $B = (P, Q, R)A$ for isometries of dimensions a, c, b , where

$$((P, Q, R)A)_{ijk} = \sum_{i', j', k'} P_{ii'} Q_{jj'} R_{kk'} A_{i'j'k'}. \quad (1.2)$$

In characteristic two, (1.1) is the group preserving the standard symmetric bilinear form. A theorem for this group is not automatically a theorem for arbitrary quadratic-form orthogonal groups. We keep this convention explicit.

The relevant average-case model chooses the source A uniformly and allows the target B to be arbitrary. An invariant predicate identifies eligible sources. For each eligible source, the algorithm must answer every comparison correctly. On an ineligible source it returns an explicit unavailable value, denoted $\perp$. In particular, $\perp$ is neither a negative answer nor an error probability on a recognized input.

Li, Li, Qiao, Tao, and Wang pose the finite-field orthogonal and unitary question as Problem 1.8 [1]. Their formulation motivates this recognizable-source interpretation. Chizewer, Everett, Mithal, and Qiao investigate the real and complex setting [2]; their use of contractions provides a useful comparison, but finite-field forms have no positive-definite spectral theory. Chen, Grochow, Qiao, Tang, and Zhang study reductions among classical-group tensor actions [5]. Such reductions provide complexity context; they do not automatically preserve a uniform input distribution.

1.2 Three regimes, two mathematical mechanisms

The construction treats three regimes. If b is below a fixed cutoff, both small factors have bounded dimension. Keep their matrices as unknowns and eliminate the large factor by a kernel-and-Gram criterion. The result is a fixed-format equation problem, even if c is enormous.

If $a \geq 2$ and b is large, compute two families of tensor contractions. When they generate their full matrix algebras, linear intertwining equations recover the first two transformations up to scalars. A final kernel-and-Gram test handles the remaining factor. Probability enters only to show that this sufficient condition usually holds.

If $a = 1$ and b is large, use a separate matrix-boundary method. A squarefree, invertible Gram operator permits deterministic cyclic-vector construction and exact isometry tests. This branch supplies a weaker but uniform coverage bound.

```
UniformComparison(field, A, B):
  parse the input; sort the three factor dimensions a <= b <= c
  if b < B0:
    return BoundedMedianDecision(A, B)
  if a = 1:
    if not BoundaryGood(A): return unavailable
    return BoundaryDecision(A, B)
  if not FullAlgebraGood(A): return unavailable
  return TwoIntertwinerDecision(A, B)
```

The algorithms in this description are deterministic. The random choice is in the source distribution used to assess coverage. The final formal specification allows a coin machine, but a deterministic implementation can satisfy it by using the same transition for either coin value.

1.3 Statements and scope

Theorem 1.1 (Exact comparison on recognized sources).

There is an explicit, invariant predicate G , decidable in deterministic polynomial time, with the following property. For every source $A \in K^a \otimes K^c \otimes K^b$ satisfying $G(A)$, equivalence to every target B of the same format is decidable in deterministic time polynomial in abc and $\log q$. The criterion applies in every characteristic and imposes no rank hypothesis.

Sections 3 and 4 give the predicate and prove the theorem. Section 5 supplies a deterministic decision construction on every source of bounded median dimension. Its constants depend on the fixed cutoff and may be very large.

Let $\gamma_{a,b,c}(q)$ be the fraction of sources accepted by the combined recognizer. The all-format conclusion pursued in the accompanying development is one uniform recognizer and comparison machine, with positive absolute constants c_0, C , an integer cutoff B_0 , and absolute polynomial runtime constants, such that

$$\begin{aligned} \gamma_{a,b,c}(q) &\geq c_0/q && \text{for every positive format,} \\ 1 - \gamma_{a,b,c}(q) &\leq \min\{1, C(1 + \log b)/\sqrt{b}\} && \text{if } a \geq 2, \\ \gamma_{a,b,c}(q) &= 1 && \text{if } b < B_0. \end{aligned} \tag{1.3}$$

Every recognized source must admit exact comparison to every target in time polynomial in $(abc + 1)(1 + \log q)$, with constants chosen before the field or dimensions. Section 9 explains the mathematical assembly of these conclusions.

This manuscript is an account of that construction, not a declaration that the final Lean theorem has been certified. Short central arguments are proved here; the longer primary-counting and geometric foundations are summarized and assigned explicit sources in Appendix B. The current ledger records their mathematical component endpoints as closed, while the combined machine implementation and final audit remain unfinished. No novelty claim is made for classical ingredients such as Witt extension, finite-field norm surjectivity, or cyclic-vector construction.

1.4 How to read the paper

The central comparison mechanism is in Sections 2-4. Readers primarily interested in the algorithm can then read the pseudocode in Section 5 and the deterministic boundary construction in Section 6. Section 7 explains why the recognizable family is large and why dependence cannot be ignored. Sections 8 and 9 address the actual computational model. Section 10 distinguishes unfinished certification from broader research questions.

For a probabilistic perspective, Fulman's survey [6] is a useful introduction to primary decomposition and finite-field random matrices. Sercombe and Shalev [7] study random generation of associative algebras. Their results concern different sampling laws from some of the conditioned self-adjoint distributions needed here, so we use them as context rather than as an automatic replacement for the anchor argument.

2 Kernels, Gram matrices, and the last factor

We begin with the reconstruction principle used twice later. Regard a rectangular matrix as a linear map from its column-coordinate space. Equality of kernels makes a prescribed map between its column spans well defined; equality of Gram matrices makes that map an isometry.

Here is why both kinds of information are necessary. Over $\mathbb{F}_5$, the nonzero column $\mathbf{h} = (1, 2)^T$ has $\mathbf{h}^T \mathbf{h} = 0$, exactly as the zero column does. Their Gram matrices agree, but no invertible transformation sends one to the other. Their kernels differ. Over finite fields, a zero self-pairing need not mean a zero vector; retaining linear dependencies repairs this loss of information.

Lemma 2.1 (Unmarked recovery).

Suppose either that the action is unitary or that it is orthogonal in odd characteristic. For matrices $\mathbf{H}, \mathbf{D} \in K^{c \times m}$, there exists $\mathbf{Q} \in \mathcal{G}_c(K)$ with $\mathbf{QH} = \mathbf{D}$ if and only if

$$\ker \mathbf{H} = \ker \mathbf{D}, \quad \mathbf{H}^* \mathbf{H} = \mathbf{D}^* \mathbf{D}. \quad (2.1)$$

Proof.

Necessity follows from invertibility and preservation of the ambient form. Conversely, kernel equality defines a bijection $\mathbf{H}\mathbf{x} \mapsto \mathbf{D}\mathbf{x}$ from $\text{im } \mathbf{H}$ to $\text{im } \mathbf{D}$. The Gram identity says that this bijection preserves all pairings. Witt extension extends it to the ambient nonsingular space. The subspaces themselves need not be nonsingular; see, for example, the extension theorem in Sprehn and Wahl [4].

□

The usual formulation must be modified for the standard symmetric form in characteristic two. Let $\mathbf{e} = (1, \dots, 1)^T$. Since

$$\mathbf{x}^T \mathbf{x} = (\mathbf{x}^T \mathbf{e})^2, \quad (2.2)$$

every standard orthogonal isometry fixes $\mathbf{e}$. Forgetting this distinguished vector can turn a necessary Gram condition into an incorrect equivalence test.

For a second example, work over $\mathbb{F}_2$ in dimension four. The columns $\mathbf{e} = (1, 1, 1, 1)^T$ and $\mathbf{d} = (1, 1, 0, 0)^T$ are nonzero, have the same zero Gram matrix, and have equal kernels as maps from $\mathbb{F}_2$. Nevertheless, no standard orthogonal matrix carries $\mathbf{e}$ to $\mathbf{d}$, because every such matrix fixes $\mathbf{e}$. Appending the marker detects exactly this obstruction.

Lemma 2.2 (Marked recovery in characteristic two).

Let K be a finite field of characteristic two. There exists $\mathbf{Q} \in \mathbf{O}_I(c, K)$ with $\mathbf{QH} = \mathbf{D}$ if and only if

$$\ker[\mathbf{H} \mid \mathbf{e}] = \ker[\mathbf{D} \mid \mathbf{e}], \quad \mathbf{H}^T \mathbf{H} = \mathbf{D}^T \mathbf{D}. \quad (2.3)$$

Proof.

Necessity includes the fact that $\mathbf{Q}\mathbf{e} = \mathbf{e}$. For sufficiency, augmented kernel equality defines an injective map on $\text{im } \mathbf{H} + K\mathbf{e}$, carrying the columns of $\mathbf{H}$ to those of $\mathbf{D}$ and fixing $\mathbf{e}$. Gram equality preserves the column pairings. Its diagonal entries, together with (2.2) and injectivity of squaring, preserve the pairings with $\mathbf{e}$. Thus this map is an isometry.

Here is a direct extension argument in this particular form. Suppose an isometry $\mathbf{f} : U \rightarrow U'$ fixes $\mathbf{e}$, with $\mathbf{e} \in U$, and choose $\mathbf{x} \notin U$. Seek a vector $\mathbf{y}$ satisfying $\langle \mathbf{y}, \mathbf{f}(\mathbf{u}) \rangle = \langle \mathbf{x}, \mathbf{u} \rangle$ for every $\mathbf{u} \in U$. Nonsingularity makes each of the two ambient pairing maps onto the dual of its subspace surjective, so corresponding fibers have the same cardinality. The parts of those fibers lying in U and U' correspond under $\mathbf{f}$. The first fiber contains $\mathbf{x} \notin U$; consequently the second contains some $\mathbf{y} \notin U'$. The equation with $\mathbf{u} = \mathbf{e}$ forces $\langle \mathbf{y}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{x} \rangle$. Sending $\mathbf{x}$ to $\mathbf{y}$ therefore extends the isometry by one dimension. Repetition proves the lemma.

□

Both criteria are tested by Gaussian elimination and matrix multiplication. They decide existence of the final factor without constructing it, which is sufficient for the decision problem considered here.

3 An invariant source recognizer

Write a tensor as A_{ijk} , with index ranges a, c, b . Let $F \in K^{c \times ab}$ be its middle flattening, and let $B_i = (A_{ijk})_{j,k} \in K^{c \times b}$. We also use the slices $A_k = (A_{ijk})_{i,j} \in K^{a \times c}$. Define

$$S = FF^*, \quad R_t = \sum_{i=1}^a B_i^T \overline{S^t} \overline{B_i} \quad (0 \leq t < c), \quad Z_{k\ell} = A_k A_\ell^*. \quad (3.1)$$

The transpose on B_i is intentional. In the unitary case, replacing it by an adjoint changes the covariance law. Each R_t is self-adjoint, and $Z_{k\ell}^* = Z_{\ell k}$.

Definition 3.1.

The predicate $G(A)$ holds when

$$K[R_0, \dots, R_{c-1}] = \text{Mat}_b(K), \quad K[Z_{k\ell} : 1 \leq k, \ell \leq b] = \text{Mat}_a(K). \quad (3.2)$$

Here brackets denote the unital algebra generated by the displayed matrices.

This condition has a straightforward algorithm. Start with the identity matrix and retain a basis of its span. Multiply each newly retained matrix on the left by every generator, adding a product precisely when it increases the span. There can be at most n^2 additions in $\text{Mat}_n(K)$. When the queue is empty, its span contains every word in the generators and hence is exactly the generated algebra.

Proposition 3.2.

The predicate G is invariant under the tensor action and is decidable in polynomial time.

Proof.

Let $A' = (P, Q, R)A$, where the three matrices are isometries of dimensions a, c, b . Flattening gives $S(A') = QS(A)Q^*$. Substitution in (3.1), using $Q^T \bar{Q} = I$ and $PP^* = I$, gives

$$R_t(A') = RR_t(A)R^*. \quad (3.3)$$

The cross-Gram list transforms by conjugation by P , together with the invertible change of labels having coefficient matrix $R \otimes \bar{R}$. Its span therefore generates the conjugate algebra. Both equalities in (3.2) are invariant.

Successive powers of S , the contractions, and the basis-closure algorithm use polynomially many field operations. Gaussian elimination implements every span test. With the represented finite-field arithmetic discussed in Section 8, this gives the asserted bit bound.

□

One can think of the contraction lists as coordinate-sensitive fingerprints: equivalent tensors produce conjugate lists. The test does more than compare fingerprints, however. Full algebra generation guarantees that a possible matching map has almost no freedom, which is what turns a necessary invariant into an exact decision method.

Full algebra generation is stronger than having scalar commutant. For example, the upper-triangular 2×2 algebra has only scalar matrices in its commutant, but preserves the first coordinate line. The proof below needs the full algebra, not merely a scalar commutant.

4 Reconstruction from two intertwiner lines

Lemma 4.1.

Suppose the matrices M_j generate $\text{Mat}_n(K)$. The solution space of $XM_j = N_jX$ for all j is either zero or a one-dimensional space whose nonzero elements are invertible. If both lists have matching adjoint mates, every such element satisfies $X^*X = dI$ for a nonzero scalar d .

Proof.

The kernel of a solution is invariant under every M_j , hence under every matrix. A nonzero solution is therefore injective and thus invertible. For two nonzero solutions X, Y , the matrix $X^{-1}Y$ commutes with the generated algebra. Commuting with the matrix units forces it to be scalar. Finally, take the adjoint of the equation indexed by the mate of j . This gives $M_jX^* = X^*N_j$, so X^*X commutes with each M_j and is scalar. Its scalar is nonzero because X is invertible. $\square$

In the unitary case d belongs to the fixed field. A scalar multiple of X is an isometry exactly when $\alpha\bar{\alpha} = d^{-1}$ is solvable. Over an odd orthogonal field this is the square test $d^{(q-1)/2} = 1$. In characteristic two every nonzero scalar is a square. In the unitary case the norm map $K^\times \rightarrow \mathbb{F}_q^\times$ is surjective, since its image in the cyclic group $K^\times$ has order $q-1$. These are existence tests; except for the characteristic-two marker, roots will not be constructed.

Let A be recognized and B arbitrary. Solve the first linear system

$$XR_t(A) = R_t(B)X \quad (0 \leq t < c). \quad (4.1)$$

Reject if the solution space is zero. Otherwise choose a nonzero solution and write $X^*X = d_bI$. Reject if it fails the applicable scalar-normalization test.

Apply X^{-1} to the b -factor of B , obtaining slices Y_k . The second target list is

$$N_{kl} = d_b Y_k Y_l^*. \quad (4.2)$$

The factor d_b is necessary because X has not been normalized to an isometry. Solve

$$P_0 Z_{kl}(A) = N_{kl} P_0. \quad (4.3)$$

Again reject the zero solution space; otherwise write $P_0^* P_0 = d_a I$ and perform the scalar-normalization test.

Let $C = (P_0, I, X)A$, let $H, D \in K^{c \times ab}$ be the middle flattenings of C, B , and put $s = (d_a d_b)^{-1}$. In the unitary and odd orthogonal cases the final test is

$$\ker H = \ker D, \quad D^* D = s H^* H. \quad (4.4)$$

In characteristic two, calculate $\lambda = s^{q/2}$, so that $\lambda^2 = s$, and replace the kernel test by

$$\ker[\lambda H \mid e] = \ker[D \mid e]. \quad (4.5)$$

Proof of Theorem 1.1.

We first check completeness. If $B = (P, Q, R)A$, covariance shows that R solves (4.1). Lemma 4.1 gives $X = uR$ for some nonzero u , so $d_b = u\bar{u}$. After applying X^{-1} , we have $Y_k = u^{-1}PA_kQ^T$. Equation (4.2) consequently becomes $N_{kl} = PZ_{kl}P^*$. Thus $P_0 = vP$, for a nonzero v , and $d_a = v\bar{v}$. Neither scalar test can reject an equivalent target. The flattenings satisfy $D = (uv)^{-1}QH$, proving (4.4). In characteristic two the unique square root λ equals $(uv)^{-1}$, which proves the marked test as well.

For soundness, the passed scalar tests give scalars α, β with $\alpha\bar{\alpha} = d_a^{-1}$ and $\beta\bar{\beta} = d_b^{-1}$. Thus αP_0 and βX are isometries. Set $\lambda = \alpha\beta$. Multiplication by $\lambda \neq 0$ preserves the kernel of H , and $(\lambda H)^*(\lambda H) = sH^*H$. Lemma 2.1, or Lemma 2.2 with the uniquely determined λ , supplies an isometry Q satisfying $Q\lambda H = D$. Consequently $B = (\alpha P_0, Q, \beta X)A$.

The computation uses the polynomial-size algebra tests, two linear systems with at most b^2 and a^2 unknowns, matrix inversion, and kernel and Gram comparisons. Binary powering implements the scalar tests. None of these steps searches through the field or factors a polynomial. Their combined bit cost is polynomial. □

Algorithmic summary.

The following concise version emphasizes that a failure of recognition is different from a negative equivalence answer.

```
DecideOnRecognizedSource(A, B):
  if not G(A): return unavailable
  compute the two contraction lists for A and B
  solve the first intertwiner system; reject if it is zero
  test the first scalar normalization
  correct the target cross-Grams by the scalar db
  solve the second intertwiner system; reject if it is zero
  test the second scalar normalization
  return the final Gram and (marked) kernel comparison
```

5 The bounded-median branch

When $a \leq b \leq B$, it is unnecessary to recover factors from generating algebras. Instead, keep the two small factors as unknowns and eliminate the large one by Section 2. This works for every input tensor, including singular and zero tensors.

Proposition 5.1 (Bounded polynomial encoding).

For fixed B , equivalence in every format with median at most B reduces, by polynomial-time preprocessing, to rational feasibility of equations of degree at most four in at most $4B^2$ scalar variables over a represented finite field.

Proof.

Let $F, D \in K^{c \times ab}$ be the flattenings and introduce unknown isometries $P \in \text{Mat}_a(K)$, $R \in \text{Mat}_b(K)$. Put $T = P \otimes R$ and $H = FT^T$. Impose $P^*P = I$, $R^*R = I$, and $H^*H = D^*D$.

For the orthogonal case take $\mathbf{e} = \mathbf{0}$ in odd characteristic and the all-ones vector in characteristic two. Compare the constant ranks of $[\mathbf{F} \mid \mathbf{e}]$ and $[\mathbf{D} \mid \mathbf{e}]$; unequal ranks force rejection. Otherwise compute a basis $\mathbf{z}_j$ of $\ker[\mathbf{D} \mid \mathbf{e}]$, and add $[\mathbf{H} \mid \mathbf{e}]\mathbf{z}_j = \mathbf{0}$. Since

$$[\mathbf{H} \mid \mathbf{e}] = [\mathbf{F} \mid \mathbf{e}] \text{diag}(\mathbf{T}^T, \mathbf{1}), \quad (5.1)$$

the unknown isometry equations force the left matrix to have the same rank as $[\mathbf{F} \mid \mathbf{e}]$. Kernel inclusion therefore becomes equality by rank-nullity. Recovery is now exact. In the unitary case use $\ker \mathbf{D}$ without the marker and compare $\text{rank } \mathbf{F}$ with $\text{rank } \mathbf{D}$.

The entries of $\mathbf{H}$ have degree two in the unknown entries of $\mathbf{P}, \mathbf{R}$; its Gram equations have degree four. In the unitary case express each unknown in a quadratic basis over the fixed field. Conjugation is then linear and multiplication bilinear, so the degree bound is unchanged. There are at most $2(a^2 + b^2) \leq 4B^2$ fixed-field variables.

Finally, replace the equations by a basis of their span as formal coefficient vectors. This leaves at most $\binom{4B^2+4}{4}$ equations and preserves their common zeros over every extension field. No independence assertion about polynomial functions on a small finite field is involved.

□

For perspective, a format with $\mathbf{a} = 2$, $\mathbf{b} = 3$, and arbitrarily large $\mathbf{c}$ needs at most $2(2^2 + 3^2) = 26$ fixed-field unknowns in the unitary encoding. Reading the long tensor still costs time, but its long axis does not create more unknown transformation entries. This is why the median dimension is the useful parameter.

5.1 A fixed-format rational feasibility procedure

The required backend concerns an arbitrary affine zero set, not just a finite one. We give the construction because replacing it by the words “solve the polynomial system” would omit the principal difficulty.

Theorem 5.2 (Fixed-format finite-field decision).

For fixed numbers $\mathbf{n}, \mathbf{d}$, rational nonemptiness of a polynomial system of degree at most $\mathbf{d}$ in $\mathbf{n}$ variables over a represented finite field $\mathbf{k}$ is deterministically decidable in time polynomial in the input length and $\log |\mathbf{k}|$. The constants may depend on $\mathbf{n}, \mathbf{d}$. The construction uses effective algebraic-closure elimination, projective degree bounds, and the finite-field point estimate stated below.

We supply the construction and its correctness argument at the usual algebraic-geometric level. The projective degree theory and the cited point estimate are mathematical dependencies, not new hypotheses on the input. Their complete Lean realization is not asserted here.

First compress the equations. If $s \leq \binom{n+d}{n}$ equations remain, the affine Bezout inequality bounds the sum of the degrees of the geometric components by $\Delta = \max(1, d)^s$. We also use the standard degree-separation fact: an affine irreducible variety of degree δ is the common zero set of the polynomials of degree at most δ which vanish on it. To see the geometric content, separate an exterior point from the projective closure by a linear projection to a hypersurface. Choose its center disjoint from the closure and its fiber through the point disjoint from the closure. The projected hypersurface has degree at most δ ; pulling back its equation gives the separator. These are the degree-theoretic inputs to the construction.

Fix a monomial basis through degree Δ . A component is encoded by the unique reduced-row-echelon basis of its truncated vanishing space. There are finitely many pivot patterns, each with finitely many free coefficient positions. For every pattern, construct a first-order polynomial formula saying that the

represented set is nonempty, its rows form its entire truncated vanishing space, it is irreducible, and it is maximal among such subsets of the input zero set. These conditions have bounded syntax. Appendix A explains why irreducibility and maximality require no unbounded quantification over polynomials.

The union of the resulting parameter sets is finite and is in bijection with the geometric components. Frobenius acts on their coefficients, and uniqueness of reduced row echelon form implies that a parameter is rational exactly when its component is individually Frobenius-fixed. Algebraic-closure quantifier elimination converts the parameter formulas into finitely many clauses of equations and inequations. A clause with inequation product $g \neq 0$ becomes a closed system by adjoining $wg - 1 = 0$. This preserves both geometric and rational points bijectively.

Lemma 5.3 (Finite-algebra point test).

Let $A = k[t_1, \dots, t_N]/I$ be finite-dimensional, and let $Q = |k|$. If J is the ideal of A generated by the classes $t_i^Q - t_i$, then

$$\#V(I)(k) = \dim_k A - \dim_k J. \quad (5.2)$$

Proof.

The polynomial ring modulo all field equations is the product of one copy of k for each point of k^N . Quotienting further by I retains precisely the factors at which every member of I vanishes. Thus A/J is a product indexed by $V(I)(k)$, and dimension gives the identity. If $b_1, \dots, b_h$ is a basis of A , the vectors $b_j(t_i^Q - t_i)$ span J ; its dimension is a matrix rank. Binary powering computes these defects inside A , without constructing degree- Q polynomials. □

For a positive-dimensional, geometrically irreducible affine variety defined over k , a uniform point-existence threshold follows from Cafure and Matera [3, Theorem 7.1]. If its dimension is $r > 0$, its degree is δ , and $Q > 2(r + 1)\delta^2$, their estimate is

$$|\#V(k) - Q^r| < (\delta - 1)(\delta - 2)Q^{r-1/2} + 5\delta^{13/3}Q^{r-1}. \quad (5.3)$$

For example, $Q > 2(n + 1)\Delta^2 + 4\Delta^4 + 20\Delta^5 + 1$ makes the right-hand side smaller than Q^r for $r \leq n$, $\delta \leq \Delta$. A zero-dimensional Frobenius-fixed geometric component is already a rational point.

Suppose, on the other hand, that no component is individually fixed by Frobenius. Every rational point lies in a component and its distinct conjugate. It therefore lies in the collision locus

$$C(X) = \bigcup_{i < j} (X_i \cap X_j), \quad X(k) = C(X)(k). \quad (5.4)$$

Distinct maximal components do not contain one another. Each intersection has smaller dimension, even if the component dimensions differ. A formula for $C(X)$ quantifies over two distinct canonical component parameters. Eliminate those quantifiers and again replace inequations by inverse-variable equations. Every resulting child has lower dimension. Taking a closure instead would be incorrect: it could introduce rational points excluded by the inequations.

Proof of Theorem 5.2.

Effective elimination bounds give computable bounds on all child formats. Iterate these format bounds for $n + 1$ levels and take Q_* larger than every point-existence threshold encountered. This is a constant depending only on n, d .

If $Q \leq Q_*$, direct enumeration of k^n is a parameter-dependent constant amount of work. Otherwise first reject geometrically empty input, using the unit-ideal test. Test the finite component-parameter sets with Lemma 5.3. A rational parameter proves existence of a rational point by (5.3), or by the zero-dimensional observation. If none exists, recurse on the exact collision children. Dimension strictly decreases, so the recursion has at most $n + 1$ levels; its branching and symbolic sizes have parameter-only bounds. Equation (5.4) proves correctness of each recursive disjunction. The only unbounded field-size dependence is in arithmetic and binary Frobenius powering, both polynomial in $\log Q$.

□

What the solver actually does.

The geometric discussion translates into the following finite procedure. The format bounds and all necessary templates are fixed before reading field coefficients. The threshold Q_* dominates the thresholds of every possible descendant, not merely that of the initial system.

```
FiniteFieldFeasible(equations, format):
    compress the coefficient rows
    if field_size <= threshold_for_all_descendants(format):
        enumerate field assignments and test the equations
    if the zero set over the algebraic closure is empty:
        return NO
    if the finite canonical component-parameter set has a field point:
        return YES
    construct the exact collision clauses
    replace each inequation  $g \neq 0$  by a fresh equation  $w \cdot g = 1$ 
    return OR(FiniteFieldFeasible(child) for every collision child)
```

For intuition, over $\mathbb{F}_q$ take an element $\alpha \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$. The two conjugate lines $y = \alpha x$ and $y = \alpha^q x$ have a union defined over $\mathbb{F}_q$. Neither line is individually defined there. Any rational point lies on both, hence at their intersection $(0, 0)$. Collision descent retains that point instead of incorrectly rejecting the union because its individual lines are not rational.

The finite-algebra calculation also tolerates repeated roots. For example $k[t]/(t^2)$ has dimension two, but its zero set consists of one point. Imposing $t^Q - t = 0$ kills the nilpotent class of t , leaving dimension one, as (5.2) predicts. Counting quotient dimension before imposing the field equations would be wrong.

Corollary 5.4.

For every fixed B , orthogonal and unitary tensor equivalence with median dimension at most B admits a deterministic polynomial-bit-time decision algorithm on all inputs.

Proof.

Apply Theorem 5.2 to Proposition 5.1.

□

This is a decision theorem, not a claim that a rational point or an isometry witness can be produced by the same procedure. Its large-field acceptance step deliberately uses existence rather than witness extraction.

6 The matrix boundary

When one tensor dimension is one, its scalar isometry can be absorbed into another factor. The problem becomes equivalence of rectangular matrices $F, D \in K^{c \times b}$ under left and right isometries. The convenient source condition is

$$\det(F^*F) \neq 0, \quad \chi_{F^*F} \text{ is squarefree.} \quad (6.1)$$

It is invariant and is recognized by a determinant, a characteristic polynomial, and a polynomial gcd. Invertibility of the Gram matrix makes the image nonsingular; squarefreeness makes the operator cyclic.

6.1 Unitary and characteristic-two decisions

In the unitary case, two squarefree self-adjoint operators with the same characteristic polynomial are isometrically conjugate. One way to prove this is to decompose their etale algebras into primary fields. The involution either preserves a field factor, where norm surjectivity identifies the one-dimensional Hermitian forms, or exchanges two factors, where a choice on one determines the other. Orthogonal sums assemble the local isometries. Thus, on sources satisfying (6.1), the matrix test is simply that the target also satisfies (6.1) and that the two Gram characteristic polynomials agree. Unmarked recovery supplies the left factor.

In characteristic two, let $K(S, v) = [v, Sv, \dots, S^{b-1}v]$ be the Krylov matrix. For a standard symmetric operator with squarefree characteristic polynomial, the all-ones vector is cyclic. Indeed, the simple primary decomposition is perpendicular. On each primary field the nondegenerate balanced form is a nonzero trace pairing. In characteristic two it is nonalternating: squaring is bijective, so its diagonal trace functional cannot vanish identically. The projection of the canonical vector onto each primary summand is consequently nonzero, and generates that simple summand.

Put $S_F = F^T F$, $S_D = D^T D$, and calculate

$$Q = K(S_F, e_b)K(S_D, e_b)^{-1}. \quad (6.2)$$

Reject if the second Krylov matrix is singular. Otherwise test $Q^T Q = I$, $Q^T S_F Q = S_D$, and equality of the kernels of $[FQ \mid e_c]$ and $[D \mid e_c]$. Any genuine right isometry fixes e_b , so its values on the Krylov basis force it to equal Q . Marked recovery proves sufficiency. There is no randomized search in this case.

6.2 Odd-characteristic decision without factorization

Assume both Gram matrices satisfy (6.1) and have common characteristic polynomial f . Construct cyclic vectors v, w for S_D, S_F , respectively, and put

$$T = K(S_F, w)K(S_D, v)^{-1}, \quad D_0 = T^T T. \quad (6.3)$$

The first matrix intertwines the Gram operators; the second commutes with S_D . Cyclicity identifies that commutant with the etale algebra $K[x]/(f)$. Correcting T to an isometry is equivalent to taking a square root of D_0^{-1} in this algebra, and hence to testing squareness in each primary field. The inverse has the same square class as the original element.

These tests do not require factoring f . For $1 \leq d \leq b$, calculate

$$g_d = \gcd(f, x^{q^d} - x), \quad \ker g_d(S_D) \subseteq \ker(D_0^{(q^d-1)/2} - I). \quad (6.4)$$

Every irreducible factor of degree e appears at $d = e$, where the condition is precisely the finite-field square test. If it is square in its own field, it passes all tests in the extension fields for which it reappears. Thus the conjunction of (6.4) is necessary and sufficient. Compute x^{q^d} modulo f by binary powering; its unreduced degree must never be used as an array size.

6.3 A deterministic cyclic-vector construction

The choice of cyclic vectors need not be random. Deterministic methods are classical; see Augot and Camion [8]. The following slower construction is convenient because its invariant is elementary and its formal interfaces are small.

For a vector z , let a_z be the monic polynomial of least degree satisfying $a_z(T)z = 0$. Compute it by finding the first dependence in $z, Tz, T^2z, \dots$; Gaussian elimination suffices. If χ_T is squarefree, perform the following sweep over the standard basis.

```
SquarefreeCyclicVector(T):
  v := 0; f := 1
  for each standard basis vector e_i:
    g := monic_vector_annihilator(T, e_i)
    h := g / monic_gcd(f, g)
    w := f(T) * e_i          # use the old f
    v := v + w
    f := f * h
  return v
```

Lemma 6.1 (Cyclic-vector sweep).

If χ_T is squarefree, the returned vector is cyclic. The construction uses polynomially many field operations, with no enumeration of the field and no factorization of χ_T .

Proof.

After processing the first i basis vectors, maintain

$$a_v = f = \text{lcm}(a_{e_1}, \dots, a_{e_i}). \quad (6.5)$$

The annihilator of $f(T)e_{i+1}$ is $h = g / \gcd(f, g)$, by divisibility in $K[x]$. Since both f and g divide the squarefree characteristic polynomial, f and h are coprime. Bezout projectors recover v and w from $v + w$, so its annihilator is fh . This proves the invariant. At the end, the least common multiple is the minimal polynomial of T , which equals its squarefree characteristic polynomial. Its degree is the whole dimension, proving cyclicity. At most n dependence calculations are needed, each using at most n Krylov columns. Even repeated elimination gives a coarse $O(n^5)$ field-operation bound. $\square$

For a small illustration, let $T = \text{diag}(1, 2)$ over $\mathbb{F}_5$. The first basis vector contributes $x - 1$. On the second iteration, $(T - I)e_2 = e_2$, so the output is $e_1 + e_2$, with annihilator $(x - 1)(x - 2)$. The sweep combines the missing primary contributions without computing those factors in advance.

The source predicate and the square-class tests remain unchanged. Thus replacing the cyclic-vector sampler by this sweep preserves coverage and correctness while removing retry analysis from the sufficient route. Actual machine compilation still has to implement the dependence, gcd, evaluation, and loop operations.

7 Why the recognizable family is large

Exact comparison and abundance are different assertions. The previous sections give algorithms which never make an incorrect decision on an eligible source. We now explain why the tensor-derived matrices satisfy the eligibility conditions often enough. The reconstruction does not factor a characteristic polynomial or select an eigenvector. Primary factors enter only as witnesses in this probability argument.

The large-median proof uses three estimates: a spectral window theorem, a Gram-distribution transfer, and a generation estimate that remains valid after conditioning. This section gives their statements and assembly, proves the elementary cross-Gram bound, and identifies the longer counting arguments in the source notes. It is an exposition of the probability proof, not a replacement for every orbit calculation in those notes.

7.1 The cross-Gram algebra: many independent opportunities

Partition the raw slices into blocks of five, writing a block as $\mathbf{X}, \mathbf{Y}_1, \dots, \mathbf{Y}_4$. Since $\mathbf{a} \leq \mathbf{c}$, a uniform $\mathbf{a} \times \mathbf{c}$ matrix has full row rank with probability at least $1/4$. Conditional on such an $\mathbf{X}$, each map $\mathbf{Y}_j \mapsto \mathbf{X}\mathbf{Y}_j^*$ is surjective with equal fibers. Its four outputs are independent uniform matrices in $\text{Mat}_a(K)$.

Four uniform matrices in $\text{Mat}_a(K)$, for $\mathbf{a} \geq 2$, generate the full algebra with probability at least $3/8$. Here is a sufficient coarse argument. A proper generated algebra lies in a maximal proper unital subalgebra. Reducible ones stabilize a proper r -dimensional subspace. Writing $Q = |K|$, the number of such subspaces is at most $4Q^{r(a-r)}$, while four matrices preserve a specified subspace with probability $Q^{-4r(a-r)}$. Their total contribution is at most

$$4 \sum_{r=1}^{a-1} Q^{-3r(a-r)} \leq \frac{1}{2}. \quad (7.1)$$

For an irreducible proper algebra, finite-dimensional density and the little Wedderburn theorem identify it with an endomorphism algebra over a proper extension field, of dimension at most $\mathbf{a}^2/2$. For each maximal such algebra, choose a nonscalar commuting element; its centralizer is exactly that algebra by maximality. There are at most $Q^{\mathbf{a}^2}$ possibilities. Four uniform samples lie in their union with probability at most $Q^{-\mathbf{a}^2} \leq 1/16$. This leaves success at least $7/16$, and the weaker $3/8$ suffices.

A five-slice block therefore succeeds with probability at least $3/32$. Independent blocks give

$$\mathbb{P}(K[Z_{kl}] \neq \text{Mat}_a(K)) \leq (29/32)^{\lfloor b/5 \rfloor}. \quad (7.2)$$

The event is defined in raw coordinates. We do not condition it on success of the other reconstruction stage.

7.2 The spectral window: a useful block rather than an eigenvector

The first algebra in (3.2) is harder. Its generators are all contractions of powers of one middle Gram operator. A useful witness is a simple, nonzero primary factor: an irreducible factor occurring exactly once in the characteristic polynomial. Its primary subspace is nonsingular, its operator is invertible, and its projector is a polynomial in that operator.

Consider a uniform standard symmetric or Hermitian self-adjoint operator of dimension $n \geq b$. Let Z_h indicate simple occurrence of an irreducible polynomial h of degree d over $\mathbb{F}_q$. The comparison weights are

$$w_h = q^{-d} \quad (\text{symmetric}), \quad w_h = (q^d - (-1)^d)^{-1} \quad (\text{Hermitian}). \quad (7.3)$$

For Hermitian operators the characteristic polynomial belongs to the fixed field. Odd-degree factors remain irreducible over K , whereas even-degree factors split into two conjugate pieces whose combined block is nonsingular. This is the origin of the parity in the second weight.

Choose a lower degree cutoff D of order $\log b$, an upper cutoff L of order $b/\log b$, and an even truncation order k of order $\log b$. For distinct eligible factors in a set J with $|J| \leq k$, the joint estimate has the form

$$\left| \mathbb{E} \prod_{h \in J} Z_h - \prod_{h \in J} w_h \right| \leq \varepsilon \prod_{h \in J} w_h, \quad \varepsilon \leq C_1 (kq^{-D} + q^{-(n-kL-1)/2}). \quad (7.4)$$

The cutoffs ensure a large nonempty complement. The estimate comes from counting marked primary decompositions, dividing by the appropriate isometry centralizers, and bounding the probability that the complement contains any selected factor. That last bound includes repeated factors. Ordinary form types and Hermitian fixed/paired types must all be included. In characteristic two an alternating complement can occur even when the ambient symmetric form is nonalternating.

Put $Z = \sum_h Z_h$ and $\mu = \sum_h w_h$. Summing (7.4) over subsets controls the factorial moments of Z . Finite Bonferroni comparison gives

$$\mathbb{P}(Z = 0) \leq e^{-\mu} + \frac{\mu^{k+1}}{(k+1)!} + \varepsilon e^\mu. \quad (7.5)$$

For an elementary verification, the even truncation of inclusion-exclusion has excess $\binom{z-1}{k}$ over the zero-event indicator at a positive integer z ; this is at most $\binom{z}{k+1}$. Compare with independent Bernoulli variables of means w_h , whose elementary symmetric sums are at most $\mu^j/j!$. This comparison does not assert that actual primary factors are independent.

Irreducible-polynomial counts give $\mu = \log(L/D) + O(1)$. With sufficiently generous absolute constants in the cutoffs, the last two terms in (7.5) are negligible, while the first is of order D/L . Thus the chance of finding no useful block is

$$O((1 + \log b)^2/b). \quad (7.6)$$

Finite exceptional sizes are absorbed by enlarging the constant. The internal large-median proof, Sections 7.1-7.4, develops the orbit counts and normalized group-order products behind (7.4); the relevant formal packages are p15-p19. These operator ensembles must not be replaced by a uniform random polynomial model.

7.3 The tensor Gram is not a uniform random operator

Even a square random matrix does not produce a uniform Gram matrix. What is sufficient is a bounded second moment of its density. If $\mathbf{X}$ is uniform $N \times M$, $M \geq N$, and ρ is the density of $\mathbf{X}\mathbf{X}^*$ relative to the uniform symmetric or Hermitian ensemble, the transfer estimate is

$$\mathbb{E}_{\text{unif}} \rho^2 \leq C_2, \quad \mathbb{P}_{\text{Gram}}(E) \leq \sqrt{C_2 \mathbb{P}_{\text{unif}}(E)}. \quad (7.7)$$

The second inequality is Cauchy-Schwarz. The first follows by Fourier expansion: the second moment is the sum of squared character averages, and the averages decay with the rank of the test form. In characteristic two the diagonal characters require a separate calculation. For alternating spaces one uses their actual independent coordinates; the ordinary matrix-trace pairing would lose information.

Use $N = \min(c, ab)$ and $M = \max(c, ab)$. Then $N \geq b$. If the middle flattening is tall, use the opposite Gram F^*F . The identity

$$x^{ab} \det(xI_c - FF^*) = x^c \det(xI_{ab} - F^*F) \quad (7.8)$$

preserves every nonzero primary factor with multiplicity. The flattening identifies the corresponding invertible primary spaces. Applying (7.7) to (7.6) gives an exceptional probability of order $(1 + \log b)/\sqrt{b}$ for the actual tensor Gram. This explains the square root in the final estimate.

7.4 The anchor: why a block forces generation

A spectral block alone need not generate a full matrix algebra. The low-rank-anchor estimate supplies the missing step. In its normalized form, if H_0 is the Gram contribution from r independent hyperbolic coordinates, S_0 is a fixed self-adjoint shift, and T is an independent uniform operator in the relevant self-adjoint or alternating-associated space, then the project argument gives

$$\mathbb{P}(K[H_0 + S_0, T] \neq \text{Mat}_n(K)) \leq 640q^{-r/128}, \quad 256 \leq r \leq n/2. \quad (7.9)$$

The models and characteristic-specific coordinates are specified in the anchor source note. Here r measures the extracted hyperbolic contribution; it is not the tensor rank.

Two obstructions account for failure. The generated algebra may preserve a proper subspace; then count subspaces and the linear conditions needed to preserve them. Or it may act irreducibly but be an algebra over a larger field. In that case its entire commutant is a finite division ring, hence a field. Because the generated algebra is stable under adjoints, its commutant is also adjoint-stable. Counting compatible forms and extension-field embeddings bounds this second contribution. Using an arbitrary field found inside the commutant would not justify adjoint stability.

The shift is controlled by a simple fiber observation. After fixing one vector of each hyperbolic pair, the remaining dependence is additive. The inverse image of a coset of a linear event is either empty or a translate of its kernel. A fixed shift therefore cannot increase its probability beyond the zero-coset probability. This supplies uniformity in S_0 , including when the complementary reference ensemble is alternating but the shift is not. The full codimension and embedding sums are in the anchor note; (7.9) is not an application of an unrestricted random-matrix generation theorem.

7.5 Selection must come after counting

Choosing a favorable spectral block after looking at the tensor creates dependence. A proof that treats the chosen block as independent of the remaining entries would be invalid. The safe order is to fix a subspace, analyze its event, sum over all subspaces, and only then use spectral existence.

Fix a nonsingular d -subspace W of the middle space. In adapted coordinates write the flattening as rows M, N , with M having d rows. If M has full row rank, invariance of W is exactly $NM^* = 0$, an event of conditional probability

$$|K|^{-d(c-d)}. \quad (7.10)$$

Conditional on it, the rows of N are independent uniform vectors in the perpendicular of the row space of M . They are not unrestricted uniform vectors.

The conditioned-Gram estimate replaces the complementary contraction by an independent uniform operator, with averaged total-variation error at most

$$C_3 \sqrt{b} q^{-b/16} \quad (7.11)$$

when $d \leq b/64$ and b exceeds an absolute threshold. The averaging is over the same full-rank anchor data; it is not a worst-case assertion for every subspace. Its proof combines subspace-intersection counts with rank-dependent Fourier estimates, retaining the characteristic-two diagonal terms.

Write H_W for the contraction contributed by W , and T_W for its complement, so that $R_0 = H_W + T_W$. Combining (7.9) and (7.11) bounds the bad-generation event for this fixed block by

$$|K|^{-d(c-d)} \left(640 q^{-r/128} + C_3 \sqrt{b} q^{-b/16} \right). \quad (7.12)$$

The column form permits a suitable r growing with the lower spectral cutoff. There are at most $4|K|^{d(c-d)}$ subspaces of dimension d . The large factors cancel when we sum (7.12); choosing the lower cutoff as a sufficiently large multiple of $\log b$ makes the residual contribution smaller than the spectral error.

On spectral success the block projector is $p(\mathcal{S})$ for a polynomial of degree below c . Its contraction is therefore a linear combination of the R_t . Generation by H_W, R_0 implies generation by the whole first list. Combining this observation with (7.2) gives

$$\mathbb{P}[-G(A)] \leq \min\{1, C_{\text{large}}(1 + \log b)/\sqrt{b}\}, \quad a \geq 2. \quad (7.13)$$

The probability proof uses primary blocks to certify existence, whereas the recognizer computes the generated algebras directly. No spectral-block finder is hidden in the algorithm.

7.6 Boundary coverage

The matrix source predicate (6.1) requires its own counting argument for actual rectangular Grams. The assembled boundary endpoint recorded in the project gives an absolute $c_{\text{bdry}} > 0$ such that the predicate has density at least c_{bdry}/q , in both actions and all dimension-one placements in the range assigned to that branch. Small formats are handled by the bounded-median branch.

The counting proceeds through regular squarefree self-adjoint operators and their embeddings as rectangular Grams. Counting operators alone would omit the embedding weights. The formal boundary-density package p25 records the completed mathematical endpoint. This $1/q$ guarantee is different from

the density tending to one in (7.13); the latter applies only when every dimension is at least two.

8 Representations, bit cost, and uniformity

A field is supplied in a polynomial-quotient presentation $\mathbb{F}_p[t]/(\mu)$, with a monic irreducible modulus and coordinate arrays. Addition is coordinatewise, multiplication is convolution followed by reduction, and inversion may use binary powering to $|K| - 2$. These operations have polynomial bit cost in $\log |K|$. Field size is not a unit-cost input, and enumeration of q^r assignments is not polynomial in $\log q$ when $r > 0$.

For the unitary bounded-median encoding, work through the fixed field of the map $x \mapsto x^q$. The implementation can retain ambient-field encodings and test fixedness by $x^q = x$; it need not construct a second power-basis presentation. If θ is the supplied power-basis generator of K , it is outside the proper fixed field, so $1, \theta$ is a quadratic basis. With $s = \theta + \bar{\theta}$ and $t = -\theta\bar{\theta}$, one has

$$\theta^2 = s\theta + t, \quad \overline{u + v\theta} = u + sv - v\theta. \quad (8.1)$$

Thus conjugation of unknowns is linear over the fixed field. No random search for a field generator is necessary.

For the recognized large-tensor branch, the powers of the middle Gram, matrix-algebra closure, and intertwiner systems all have polynomial operation counts. For the odd matrix boundary, Lemma 6.1 supplies cyclic vectors deterministically. All polynomial powers in its subsequent square-class tests are computed modulo bounded-degree polynomials or through bounded-size matrix products.

An operational proof must also address arbitrary input bit strings. Before allocating a matrix, the parser checks dimensions and coordinate counts against the available string length. Arithmetic routines are given bounded behavior on malformed inputs. Correctness is asserted on valid field presentations, but polynomial-time recognition must be a total statement. Finally, field operations, matrix routines, branching, and the fixed-format solver must be composed into one actual machine with a uniform bound. Existing compiler theorems help with composition; they do not supply missing subroutine bounds.

9 Assembling one uniform algorithm

Choose $B_0 \geq 4$ above the finitely many exceptional thresholds and large enough that the right side of (7.13) is at most $1/2$ for all $b \geq B_0$. This is possible because $(1 + \log b)/\sqrt{b}$ tends to zero. The cutoff is fixed once; it is not selected from the input field.

Accept every source with $b < B_0$. For larger formats with $a = 1$, use (6.1); for the rest use G . Each predicate is invariant. Sorting merely reindexes factors and preserves uniform counting measure, so the construction applies to every original orientation, including ties.

The bounded branch has density one. The large branch has density at least $1/2$, and the boundary branch has density at least c_{bdry}/q . Thus

$$c_0 = \min\{c_{\text{bdry}}, 1/2\}, \quad C = C_{\text{large}} \quad (9.1)$$

gives the coverage requirements in (1.3); the nonboundary small-median failure probability is zero. This assembly is short because the hard work lies in obtaining the branch estimates uniformly.

Correctness follows by the same case split. Every accepted small-median source is decided by Corollary 5.4; every accepted matrix source by Section 6; and every accepted large source by Theorem 1.1. Rejected sources return $\perp$. The target need not itself be sampled at random or separately accepted.

At the mathematical algorithm level, each branch has a polynomial bit bound. For fixed B_0 , only finitely many bounded-solver profiles and descendant templates are required. Take the maximum of the finitely many exponents and enlarge one coefficient to cover dispatch and parsing. Since $|K|$ is either q or q^2 , its bit length is at most twice that of q , up to a constant. This produces a single bound

$$C_{\text{time}}((abc + 1)(1 + \lceil \log_2 q \rceil))^e. \quad (9.2)$$

This is an ordinary bit-complexity assembly argument. Certifying it for the project's particular finite machine additionally requires the concrete program and refinement proofs described next.

9.1 Why the formal implementation is substantial

A field-operation count does not prove an encoded-machine bound. A subroutine may have the right mathematical output while its caller reads from the wrong array, reuses a live register, or incurs an unbounded storage cost. The formal implementation must connect the mathematical state to the physical program state at every call and loop.

For the bounded solver the most demanding composition is the collision child: evaluate a finite parameter clause, build and solve its Frobenius system, extract coordinate values, restore the caller's data, and construct the next equation rows. This sequence must preserve the original system and all recursion invariants. Whole-array save/restore lemmas and certified coordinate loops are useful components, but they do not by themselves instantiate this complete routine.

The remaining work also includes matrix and polynomial call sites, branch controllers, total recognition on malformed encodings, and one shared runtime bound. An invariant that assumes an unspecified solver or assumes its runtime cannot close the final target. The actual routines must be supplied.

For a deterministic implementation with step bound T , the formal coin-machine bridge is simple: both coin transitions execute the same instruction. All coin traces follow the same computation. The survival probability is zero after T , so every finite survival sum is at most T . This satisfies the allowed expected-time specification without introducing a randomized search.

10 Further questions

The questions below are suggested by this construction. They are research directions, not claims that each has been separately posed as an open problem in the literature. Completing the encoded Lean theorem is a concrete verification task; the other questions concern stronger algorithms or sharper mathematics.

10.1 Complete certification and a reproducible release

The immediate goal is a proof of the unchanged all-format root with concrete programs and no extra solver, branch, or cost hypotheses. A release should include a frozen dependency manifest, exact theorem statements, source-linked audit receipts, and a reproducible kernel and axiom check. This is more informative than a count of compiled helper files. A useful reusable outcome would be the verified bounded-format finite-field solver itself, with its mathematical and machine-level interfaces clearly separated.

10.2 Replace the fixed-format existence proof by practical algorithms

The geometric solver establishes a sufficient asymptotic bound, but its elimination templates and point-existence thresholds may be far too large for computation. Can the tensor-derived systems be solved by a more direct algebraic method? Their degree is at most four and their unknowns are constrained isometries, so they are not arbitrary polynomial systems. Exploiting those special features could reduce both the computational constants and the formalization burden.

10.3 Improve coverage and the median error

The large-branch bound loses a square root when transferring from a uniform operator to a Gram distribution. Can an event-specific estimate avoid that loss and improve the decay in b ? Can one recognize more boundary matrices while retaining deterministic exact comparison? The present c/q boundary guarantee need not reflect the actual largest tractable family. Neither the optimal failure rate nor the best recognizer is determined by this construction.

10.4 Recover witnesses and canonical forms

Decision does not automatically provide a transporter: explicit isometries carrying A to B . The large branch avoids constructing certain roots and the bounded solver sometimes accepts by a point-existence theorem. Turning those steps into efficient witness extraction is a substantive extension. Canonical forms are stronger still: one would need choices consistent across equivalent presentations, not merely a successful pairwise comparison.

10.5 Other forms and other input distributions

The standard bilinear orthogonal action in characteristic two should be distinguished from quadratic orthogonal actions. Extending the whole argument to prescribed nonstandard forms requires matching recovery theorems, primary counts, and conditional Gram estimates. Alternating-form calculations already present in the project are relevant ingredients, but do not supply that extension automatically.

Uniform independent entries are also a specific model. Structured tensors, planted equivalent pairs, and cryptographic instance distributions can behave differently. The source-versus-arbitrary-target correctness guarantee is useful, but transferring the coverage estimate to another distribution requires a separate argument. A decision theorem alone is not a break of a particular cryptographic construction.

10.6 Beyond the isometry action

The use of adjoints is central: it creates covariant contractions and scalar Gram corrections. Under unrestricted general linear changes, these pairings are not preserved. Extending the recognizable-source method to that setting would require new invariants or a distribution-preserving reduction. Existing worst-case reductions do not supply such a transfer. Nor does the present equivalence algorithm establish a faster matrix-multiplication exponent, preserve tensor rank under a new compression, or produce a physical quantum simulation.

For documented neighboring questions, Problems 1.4, 1.6, and 1.7 of [1] concern average-case algorithms for general tensor and trilinear-form actions, improved success fractions for four-tensor isomorphism, and more general four-tensor formats. The present construction does not resolve those questions. Any extension would have to specify both its group action and its input distribution before an average-case comparison could be made.

11 Perspective

The construction combines two complementary mechanisms. Rich contraction algebras make large tensors rigid enough for linear reconstruction, while a bounded median leaves only finitely many transformation variables and permits a geometric decision procedure. A matrix branch fills the remaining format boundary. The common recovery principle is that pairings must be accompanied by linear dependencies, with an additional marker in characteristic two.

The probability proof explains when rigidity is available; it is not needed to justify an individual recognized answer. Conversely, exact reconstruction says nothing by itself about how often recognition succeeds. Keeping these roles separate makes both the mathematics and the remaining implementation obligations visible. The current development has reached substantial component-level certification, but the final all-format machine theorem still awaits its concrete assembly and independent audit.

A Canonical component formulas and elimination

We record details behind the fixed-format solver so that its component step is not mistaken for a primary-decomposition oracle. All quantifiers in this appendix range over an algebraic closure, unless a rational-point test is explicitly invoked.

Let $\mathbf{E} = \Delta$, and encode a subspace of polynomials of degree at most $\mathbf{E}$ by a reduced-row-echelon matrix $\mathbf{U}$ with fixed pivot pattern. Let $\mathbf{Y}_{\mathbf{U}}$ be its common zero set. Require $\mathbf{Y}_{\mathbf{U}} \neq \emptyset$, and for every coefficient vector $\mathbf{v}$ of degree at most $\mathbf{E}$, impose the equivalence between vanishing of its polynomial on $\mathbf{Y}_{\mathbf{U}}$ and membership in the row span of $\mathbf{U}$. Row membership is polynomial syntax: its coefficients are the entries of $\mathbf{v}$ in the pivot positions.

The same degree bounds give at most $\mathbf{T} = \mathbf{E}^{\binom{n+\mathbf{E}}{n}}$ components of $\mathbf{Y}_{\mathbf{U}}$, each of degree at most $\mathbf{T}$. Irreducibility is equivalent to the following bounded test: for all polynomials $\mathbf{f}, \mathbf{g}$ of degree at most $\mathbf{T}^2$, if $\mathbf{fg}$ vanishes on $\mathbf{Y}_{\mathbf{U}}$, then one of $\mathbf{f}, \mathbf{g}$ vanishes there. The forward implication is immediate. For the reverse, if $\mathbf{Y}_{\mathbf{U}}$ has distinct components $\mathbf{Y}_1, \dots, \mathbf{Y}_t$, choose a degree-at-most- $\mathbf{T}$ separator vanishing on $\mathbf{Y}_1$ but not on $\mathbf{Y}_2$. For each $i \geq 2$, choose such a separator vanishing on $\mathbf{Y}_i$ but not on $\mathbf{Y}_1$. Their product is not identically zero on the irreducible set $\mathbf{Y}_1$, while its product with the first separator vanishes on the union. Both degrees fit the bound.

Require further that $\mathbf{Y}_{\mathbf{U}} \subseteq \mathbf{X}$, and that whenever it is contained in another represented nonempty canonical irreducible subset of $\mathbf{X}$, the two are equal. Every genuine component has such a bounded representation. Every passing candidate lies in a genuine component, so maximality forces equality. Canonicity then gives exactly one parameter per component. Containment is a quantified implication between polynomial equations; the construction uses no quantification over an infinite polynomial ring.

For elimination, a quantifier-free clause consists of equations $\mathbf{f}_i(\mathbf{y}, \mathbf{x}) = \mathbf{0}$ and inequations whose product is $\mathbf{g}(\mathbf{y}, \mathbf{x}) \neq \mathbf{0}$. Introduce $\mathbf{z}$ and append $\mathbf{zg} - \mathbf{1} = \mathbf{0}$. Effective Nullstellensatz gives a computable degree bound $\mathbf{H}$, depending only on the number of eliminated variables and equation degrees, for a certificate $\mathbf{1} = \sum \mathbf{h}_i \mathbf{f}_i$ whenever the specialized system is geometrically empty. Equating coefficients is a linear system $\mathbf{M}(\mathbf{y})\mathbf{c} = \mathbf{b}$.

Consistency of this linear system is expressed by the equality of the ranks of $\mathbf{M}(\mathbf{y})$ and its augmentation, using a finite Boolean combination of minors. Its negation expresses existence of a geometric solution. The construction is valid after every parameter specialization, including those which

lower degrees or annihilate coefficients. It therefore avoids the erroneous assumption that a generic Groebner basis remains valid on exceptional parameter values. Universal quantifiers are removed by negating existential ones, and elimination proceeds from the innermost subformula outward.

For completeness, a computable certificate bound can be obtained from bounded Buchberger completion. Each nonzero reduced remainder enlarges the leading-monomial ideal. Dickson's lemma prohibits an infinite sequence of such enlargements. Under the degree growth bound for each completion step, the tree of possible leading-monomial sequences is finitely branching; it has no infinite branch and is therefore finite. Exhausting this tree gives a computable parameter-only length bound. Tracking coefficient certificates through the divisions gives **H**. The resulting bound is deliberately inefficient, but its dependence is on format rather than field size.

On the finite parameter systems, standard monomials of a Groebner basis supply the quotient-algebra basis and its multiplication table. Nilpotents are retained; Lemma 5.3 removes them automatically when imposing the field equations. This completes the algorithmic description used in Section 5.

B Formalization status and source map

This snapshot was read on 2 October 2026 from the project's closure ledger and current p34 progress record. The ledger records 31 of 40 packages closed; nine remain open or partially audited. This is a statement about the recorded package contracts, not a percentage of effort and not a fresh independent replay of the complete library. The root remains open.

B.1 The exact root and the remaining packages

The declaration `AllFormatRoot` in `Problem18AllFormatRoot20260930.lean` requires one recognizer, one machine, and all uniform constants before quantifying over represented fields and positive dimensions. It includes invariant recognition, the density bounds, acceptance of every bounded-median source, exact answers on recognized sources, and the specified machine runtime. Defining this proposition in a compiling file is not a proof of it.

The open packages are p12 (remaining execution aspects of the general fixed-format solver), p33 (matrix and polynomial machine routines), p34 (the complete bounded-median machine), p35 (large and boundary machines), p36 (one total polynomial-time recognizer), p37 (one exact query machine), p38 (density for the encoded recognizer), p39 (the exact common runtime scale), and p40 (root assembly and independent audit).

These are not nine independent mathematical problems. The mathematical endpoint and field-operation count for p12 are recorded as verified; its execution work overlaps p34. The typed uniform density supplier for p38 is verified, while the connection to the final encoded recognizer remains open. A runtime-conversion theorem already supports p39. Packages p38-p40 depend on the concrete programs; they do not require inventing fresh probability or geometry.

B.2 Where the mathematical arguments live

The mathematical exposition draws on the internal research records dated 30 September 2026 listed below, the subsequent boundary reassessment, and certified component endpoints. The records are unpublished project sources. They are supplied with the companion source bundle; filenames below identify the relevant documents without assuming public repository availability.

Large-median branch. Problem18LargeMedianClosureProof20260930.md, especially Sections 2-3 for contractions and decision, Sections 6-9 for conditioning, spectral counts and selection-safe assembly, and Section 10 for the bit-complexity argument. Its older implementation-status comments are superseded by this snapshot.

Anchor estimate. Problem18AnchorInformalProof20260930.md, with the shift strengthening in Section 5 of the large-median record. This is the source of the preservation-codimension and compatible-form/field-embedding arguments behind (7.9).

Bounded-median branch and solver. Problem18BoundedMedianClosureProof20260930.md and Problem18FixedDegreeSolverFullInformalProof20260930.md, with the separate Problem18ConstructiveEliminationFullProof20260930.md. The point-existence input is identified explicitly in [3]. Appendix A gives the bounded formula and specialization-safe elimination mechanism. The current implementation uses ambient encodings for fixed-field values and does not need the older proposal for a second fixed-field presentation.

Deterministic boundary. DETERMINISTIC-BOUNDARY.md in the reassessment record dated 1 October 2026 gives the basis sweep and annihilator invariant. Its original future-tense implementation comments are historical; subsequent deterministic boundary contracts are recorded as audited. Full machine lowering remains in p35.

B.3 Representative formal endpoints

The recovery criteria are in Problem18FinalFactor20260930 and Problem18UnitaryFinalFactor20260930. The large-source Boolean correctness endpoint recorded by p31 is `rawLargeCheck_correct`. The unrestricted solver's mathematical endpoint is `unrestricted_package` in Problem18UnrestrictedSolverClosure20261001. The density supplier is `uniform_typed_density` in Problem18UniformDensityAssembly20261001.

The recent p34 record includes concrete finite-parameter decisions, Frobenius-system construction, coordinate extraction, clause evaluation, protected caller data, and bounded array-bank copying. It still lists repeated clause execution, the complete collision-child routine, recursive instantiation, physical input/output, storage bounds, and the final compiler contracts as unfinished. These entries support a precise status report; they are not a substitute for the missing root proof.

The development reuses Mathlib and project libraries. Mathlib's general role is described in [9]; importing a library does not make an arbitrary new theorem correct. Closure requires an appropriate statement, kernel-checked proof, and an axiom audit. Finite diagnostic checks can expose implementation mistakes but cannot replace a universal theorem. The source bundle records hashes for the source documents consulted in preparing this manuscript.

This draft was prepared with AI assistance. Author names and affiliations are intentionally omitted. It is a research manuscript for mathematical review, with no claim of peer review, final formal certification, or an exhaustive novelty determination.

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